Chad Jones Assignment 02 6/15/26

1. EPC is the largely centralized, node-based core for 4G LTE, while 5GC is a cloud-native, service-based, and more distributed core designed to support features like network slicing, massive IoT, and ultra-low latency.

2. Network slicing lets operators create multiple virtual networks on the same 5G infrastructure, so each use case gets tailored performance and SLAs.

3. MEC (Multi-access Edge Computing) moves compute and storage from centralized data centers to the network edge near users and base stations, cutting the physical distance data must travel and thereby reducing latency for real-time apps.

4. AI in dynamic resource allocation predicts traffic patterns and network states in real time, then automatically adjusts spectrum, power, and compute resources to maintain QoS and use infrastructure more efficiently.

5. In 5G's Service-Based Architecture, the Network Repository Function (NRF) acts as a service registry that stores and exposes information about available network functions and their services, enabling other functions to discover and select the right ones at runtime.